ARYAN GUPTA

(734) 819-3134 | guptaaryanr@gmail.com | https://www.linkedin.com/in/guptaaryanr | https://guptaaryanr.github.io/

SUMMARY

Applied physicist and engineer building ML, scientific computing, and automation systems across carbon capture, robotics, and engineering workflows. Experience spanning first-principles modeling, surrogate optimization, technical product building, and cross-functional AI pilot leadership.

PROFESSIONAL EXPERIENCE

Technip Energies – AI Solutions Initiative, Americas Region — Dec 2025 – Present *(part-time)*
(Internal initiative)
Co-Founder & Co-Lead

- Co-founded and now co-lead an Americas-wide AI initiative across 4 operating centers and 8 engineering functions, scoping and piloting 5 AI workflows for document-heavy and data-heavy EPC tasks.
- Leading development of a Python LLM-assisted natural-language-to-SQL pipeline across 3 engineering datasets that translates engineering questions into executable queries for structured retrieval, reducing lookup time from days to minutes.
- Coordinated technical development of an LLM-assisted bid tabulation workflow that parses large unstructured specification packages to extract and normalize requirement-level information for technical comparison, reducing manual review effort by ~80%.
- Building and coordinating a P&ID intelligence tool for instrument, line, loop, and cross-page connection extraction, combining object/shape detection and OCR while guiding standards-aware ML enhancements to improve detection and classification, achieving 90% accuracy of detection.

Technip Energies — Jul 2025 – Present
Specialist Engineer I – Instrumentation & Controls

- Reviewed vendor instrumentation documentation against P&IDs, PFDs, and Cause & Effect diagrams for blue hydrogen project deliverables, identifying discrepancies and supporting document alignment and issue resolution.
- Supported brownfield and carbon-capture I&C updates by reconciling legacy logic, control narratives, P&IDs, and Cause & Effect documentation within traceability and management-of-change workflows.

Technip Energies — Aug 2023 – Jul 2025,
Feb 2026 – Mar 2026 *(part-time)*
Technology Developer I (Applied Physics Lead, Novel Carbon Capture R&D Team)

- Established a first-principles molecular modeling capability for internal carbon capture R&D, developing reproducible MD/DFT pipelines across HPC and cloud that reduced simulation wall time by ~2/3 and reduced cloud spend.
- Integrated atomistic outputs into diffusion and CFD models through a nucleation-delay reformulation, improving continuum-model accuracy, reprioritizing model assumptions, and helping verify the diffusion-scale codebase.
- Evaluated multiple force-field families and benchmarked interfacial energies, diffusion rates, and crystallization barriers against experiments and literature, while also designing and supervising crystal-growth experiments.
- Built physics-informed data and ML workflows for automated parameter sweeps, experimental data ingest and processing, terabyte-scale image QA, and surrogate-assisted calibration to identify crystallization drivers and tune simulation inputs to target error bounds.
- Led uncertainty quantification and reproducibility efforts, authored 15 internal technical notes, mentored teammates, and translated results into decision-ready materials including a 70-page market analysis used in multimillion-dollar funding discussions.

HelixCases — Feb 2022 – Aug 2022
Co-Founder & Software Engineer

- Secured a $10,000 Lyft contract and built a coding competition platform used by 550+ students at 20 universities.

INDEPENDENT RESEARCH & OPEN SOFTWARE

Trajectory-Only Structural Regime Detection In Deterministic Dynamical Systems — Feb 2026 – Present
Manuscript in preparation *Code DOI: 10.5281/zenodo.19465797*

- Developing a trajectory-only framework for structural regime detection in nonlinear dynamical systems, benchmarked on FitzHugh-Nagumo and van der Pol systems with robustness, ablation, and boundary analyses.

ThermoBench-Consist (v1.0) — Sep 2025 – Oct 2025
Preprint: 10.5281/zenodo.17489426 *Code DOI: 10.5281/zenodo.17330440*

- Built a CPU-only benchmark for thermodynamic-surrogate consistency checks, combining physics-based tests, deterministic reference-grid checks, CI-backed outputs, and machine-readable one-page reports running in <30s via CLI and <60s in Colab.

REB-1: Robot Energy Benchmark (v0.1) Aug 2025 – Aug 2025
Code DOI: 10.5281/zenodo.17204853
- Built a lightweight robotics-energy benchmark with 60-second workloads, Wh→gCO_2 analysis, schema-checked outputs, and fast A/B comparison workflows.

Assessing the Limits of Graph Neural Networks for Vapor-Liquid Equilibrium May 2025 – Sep 2025
Preprint: 10.48550/arXiv.2509.10565
- Study of phase-dependent failure modes in GNN surrogates for vapor-liquid equilibrium, highlighting limits of generalization across thermodynamic regimes.

Energy-Efficient Robotics Software (2020-2024): Systematic Literature Review Jan 2025 – Aug 2025
Preprint: 10.48550/arXiv.2508.12170 *Code DOI: 10.5281/zenodo.16907564*
- Independent 79-paper systematic review on energy-efficient robotics software, with a released replication package and reporting checklist for fair energy claims.

RESEARCH EXPERIENCE

BIRDS Lab, UMich Jan 2021 – Apr 2023
Research Assistant
- Built an adaptable quadcopter hardware/software stack, implemented Wi-Fi ↔ UART bridges for real-time control/telemetry, redesigned power boards, and motor control reliability improvements in a 5-person research team.

SKILLS

- **Programming & Scientific Computing:** Python, C/C++, MATLAB/Simulink, Linux/Unix, Git, Docker, CI, Pandas, SciPy, scientific software development
- **Machine Learning & Optimization:** PyTorch, scikit-learn, OpenCV, surrogate modeling, active learning, uncertainty quantification, numerical optimization, model calibration, design of experiments, multimodal models, LLM/VLM workflows
- **Cloud, Data & Systems:** GCP, Vertex AI, MLflow, Kubernetes, Terraform, PostgreSQL, SQLAlchemy, large-scale data pipelines, data validation, experimental QA pipelines, training/inference pipelines
- **Scientific Modeling, HPC & Reproducibility:** LAMMPS, NWChem, SLURM, HPC, GPU/cloud pipelines, atomistic-to-continuum modeling, diffusion/CFD coupling, computational physics, benchmarking, reproducibility
- **Embedded & Robotics:** STM32/ARM microcontrollers, ESP32/FTDI, UART, embedded prototyping, embedded systems integration

EDUCATION

University of Michigan Sep 2019 – May 2023
BSE, Engineering Physics, *Cum Laude*
Minors: Electrical Engineering, Computer Science, Entrepreneurship